2.04 Complex Numbers: De Moivre's Theorem - Powers

Worked Solutions

1 a)

$$\begin{aligned} z^{-1} &= (\cos\theta + \mathrm{i}\sin\theta)^{-1} \\ &= \cos(-\theta) + \mathrm{i}\sin(-\theta) \\ &= \cos\theta - \mathrm{i}\sin\theta \end{aligned}$$

Alternatively:

$$\begin{aligned} z^{-1} &= \frac{1}{\cos\theta + \mathrm{i}\sin\theta} \\ &= \frac{1}{\cos\theta + \mathrm{i}\sin\theta} \times \frac{\cos\theta - \mathrm{i}\sin\theta}{\cos\theta - \mathrm{i}\sin\theta} \\ &= \frac{\cos\theta - \mathrm{i}\sin\theta}{\cos^2\theta + \sin^2\theta} \\ &= \frac{\cos\theta - \mathrm{i}\sin\theta}{1} \\ &= \cos\theta - \mathrm{i}\sin\theta \end{aligned}$$

1 b)

$$z + z^{-1} = \cos\theta + \mathrm{i}\sin\theta + \cos\theta - \mathrm{i}\sin\theta = 2\cos\theta$$

So that $2\cos\theta = z + z^{-1}$ as req.

$$z - z^{-1} = \cos\theta + \mathrm{i}\sin\theta - \cos\theta + \mathrm{i}\sin\theta = 2\mathrm{i}\sin\theta$$

So that $2\sin\theta = z - z^{-1}$ as req.

2 a)

$$\begin{aligned} z^{-n} &= (\cos\theta + \mathrm{i}\sin\theta)^{-n} \\ &= \cos(-n\theta) + \mathrm{i}\sin(-n\theta) \\ &= \cos n\theta - \mathrm{i}\sin n\theta \end{aligned}$$

2 b)

$$z + z^{-n} = \cos n\theta + \mathrm{i}\sin n\theta + \cos n\theta - \mathrm{i}\sin n\theta = 2\cos n\theta$$

So that $\cos n\theta = \frac{z^n + z^{-n}}{2}$ as req.

2 c)

$$z - z^{-1} = \cos n\theta + \mathrm{i}\sin n\theta - \cos n\theta + \mathrm{i}\sin n\theta = 2\mathrm{i}\sin n\theta$$

So that $\sin n\theta = \frac{z^n - z^{-n}}{2}$ as req.

3 a)

$$\begin{aligned}\left(z+\frac{1}{z}\right)^2 &= z^2 + 2z \times \frac{1}{z} + \frac{1}{z^2} \\ &= z^2 + 2 + \frac{1}{z^2}\end{aligned}$$

3 b)

$$\begin{aligned}\left(z+\frac{1}{z}\right)^3 &= z^3 + 3z^2 \times \frac{1}{z} + 3z \times \frac{1}{z^2} + \frac{1}{z^3} \\ &= z^3 + 3z + \frac{3}{z} + \frac{1}{z^3}\end{aligned}$$

3 c)

$$\begin{aligned}\left(z+\frac{1}{z}\right)^4 &= z^4 + 4z^3 \times \frac{1}{z} + 6z^2 \times \frac{1}{z^2} + 4z \times \frac{1}{z^3} + \frac{1}{z^4} \\ &= z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}\end{aligned}$$

3 d)

$$\begin{aligned}\left(z+\frac{1}{z}\right)^5 &= z^5 + 5z^4 \times \frac{1}{z} + 10z^3 \times \frac{1}{z^2} + 10z^2 \times \frac{1}{z^3} + 5z \times \frac{1}{z^4} + \frac{1}{z^5} \\ &= z^5 + 5z^3 + 10z + \frac{10}{z} + \frac{5}{z^3} + \frac{1}{z^5}\end{aligned}$$

3 e)

$$\begin{aligned}\left(z+\frac{1}{z}\right)^6 &= z^6 + 6z^5 \times \frac{1}{z} + 15z^4 \times \frac{1}{z^2} + 20z^3 \times \frac{1}{z^3} + 15z^2 \times \frac{1}{z^4} + 6z \times \frac{1}{z^5} + \frac{1}{z^6} \\ &= z^6 + 6z^4 + 15z^2 + 20 + \frac{15}{z^2} + \frac{6}{z^4} + \frac{1}{z^6}\end{aligned}$$

4 a)

$$\begin{aligned}\left(z-\frac{1}{z}\right)^2 &= z^2 - 2z \times \frac{1}{z} + \frac{1}{z^2} \\ &= z^2 - 2 + \frac{1}{z^2}\end{aligned}$$

4 b)

$$\begin{aligned}\left(z-\frac{1}{z}\right)^3 &= z^3 - 3z^2 \times \frac{1}{z} + 3z \times \frac{1}{z^2} - \frac{1}{z^3} \\ &= z^3 - 3z + \frac{3}{z} - \frac{1}{z^3}\end{aligned}$$

4 c)

$$\begin{aligned}\left(z-\frac{1}{z}\right)^4 &= z^4 - 4z^3 \times \frac{1}{z} + 6z^2 \times \frac{1}{z^2} - 4z \times \frac{1}{z^3} + \frac{1}{z^4} \\ &= z^4 - 4z^2 + 6 - \frac{4}{z^2} + \frac{1}{z^4}\end{aligned}$$

4 d)

$$\begin{aligned}\left(z-\frac{1}{z}\right)^5 &= z^5 - 5z^4 \times \frac{1}{z} + 10z^3 \times \frac{1}{z^2} - 10z^2 \times \frac{1}{z^3} + 5z \times \frac{1}{z^4} - \frac{1}{z^5} \\ &= z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5}\end{aligned}$$

4 e)

$$\begin{aligned}\left(z-\frac{1}{z}\right)^6 &= z^6 - 6z^5 \times \frac{1}{z} + 15z^4 \times \frac{1}{z^2} - 20z^3 \times \frac{1}{z^3} + 15z^2 \times \frac{1}{z^4} - 6z \times \frac{1}{z^5} + \frac{1}{z^6} \\ &= z^6 - 6z^4 + 15z^2 - 20 + \frac{15}{z^2} - \frac{6}{z^4} + \frac{1}{z^6}\end{aligned}$$

5 a)

Since $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ and $\mathrm{e}^{-\mathrm{i}\theta} = \cos\theta - \mathrm{i}\sin\theta$

$$\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta} = 2\cos\theta$$

So $\cos\theta = \frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})$

5 b)

Since $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ and $\mathrm{e}^{-\mathrm{i}\theta} = \cos\theta - \mathrm{i}\sin\theta$

$$\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta} = 2\mathrm{i}\sin\theta$$

So $\sin\theta = \frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})$

6 a)

$$\begin{aligned}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})^2 &= \mathrm{e}^{2\mathrm{i}\theta} + 2\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\theta} + \mathrm{e}^{-2\mathrm{i}\theta} \\ &= \mathrm{e}^{2\mathrm{i}\theta} + \mathrm{e}^{-2\mathrm{i}\theta} + 2\end{aligned}$$

6 b)

$$\begin{aligned}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})^3 &= \mathrm{e}^{3\mathrm{i}\theta}+3\mathrm{e}^{2\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\theta}+3\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-2\mathrm{i}\theta}+\mathrm{e}^{-3\mathrm{i}\theta}\\ &= \mathrm{e}^{3\mathrm{i}\theta}+\mathrm{e}^{-3\mathrm{i}\theta}+3\mathrm{e}^{\mathrm{i}\theta}+3\mathrm{e}^{-\mathrm{i}\theta}\end{aligned}$$

6 c)

$$\begin{aligned}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})^4 &= \mathrm{e}^{4\mathrm{i}\theta}+4\mathrm{e}^{3\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\theta}+6\mathrm{e}^{2\mathrm{i}\theta}\mathrm{e}^{-2\mathrm{i}\theta}+4\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-3\mathrm{i}\theta}+\mathrm{e}^{-4\mathrm{i}\theta}\\ &= \mathrm{e}^{4\mathrm{i}\theta}+\mathrm{e}^{-4\mathrm{i}\theta}+4\mathrm{e}^{2\mathrm{i}\theta}+4\mathrm{e}^{-2\mathrm{i}\theta}+6\end{aligned}$$

7 a)

$$\begin{aligned}\left[\frac{1}{\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})\right]^2 &= \left(\frac{1}{\mathrm{i}}\right)^2(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})^2\\ &= -(\mathrm{e}^{2\mathrm{i}\theta}-2\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\theta}+\mathrm{e}^{-2\mathrm{i}\theta})\\ &= 2-\mathrm{e}^{2\mathrm{i}\theta}-\mathrm{e}^{-2\mathrm{i}\theta}\end{aligned}$$

7 b)

$$\begin{aligned}\left[\frac{1}{\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})\right]^3 &= \left(\frac{1}{\mathrm{i}}\right)^3(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})^3\\ &= \mathrm{i}(\mathrm{e}^{3\mathrm{i}\theta}-3\mathrm{e}^{2\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\theta}+3\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-2\mathrm{i}\theta}-\mathrm{e}^{-3\mathrm{i}\theta})\\ &= \mathrm{i}(\mathrm{e}^{3\mathrm{i}\theta}-3\mathrm{e}^{\mathrm{i}\theta}+3\mathrm{e}^{-\mathrm{i}\theta}-\mathrm{e}^{-3\mathrm{i}\theta})\end{aligned}$$

7 c)

$$\begin{aligned}\left[\frac{1}{\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})\right]^4 &= \left(\frac{1}{\mathrm{i}}\right)^4(\mathrm{e}^{\mathrm{i}\theta}-\mathrm{e}^{-\mathrm{i}\theta})^4\\ &= \mathrm{e}^{4\mathrm{i}\theta}-4\mathrm{e}^{3\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\theta}+6\mathrm{e}^{2\mathrm{i}\theta}\mathrm{e}^{-2\mathrm{i}\theta}-4\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-3\mathrm{i}\theta}+\mathrm{e}^{-4\mathrm{i}\theta}\\ &= \mathrm{e}^{4\mathrm{i}\theta}+\mathrm{e}^{-4\mathrm{i}\theta}-4\mathrm{e}^{2\mathrm{i}\theta}-4\mathrm{e}^{-2\mathrm{i}\theta}+6\end{aligned}$$

8)

$$\begin{aligned}\left(z+\frac{1}{z}\right)^2 &= z^2+2+\frac{1}{z^2}\\ &= (z^2+\frac{1}{z^2})+2\end{aligned}$$

Since $z^n+z^{-n}=2\cos n\theta$ then

$$\left(z+\frac{1}{z}\right)^2 = 2\cos 2\theta+2$$

So $(2\cos\theta)^2 = 2\cos 2\theta+2$

$$\therefore \cos^2\theta = \frac{1}{2}\cos 2\theta+\frac{1}{2}$$

9)

$$\left(z-\frac{1}{z}\right)^2 = z^2 - 2 + \frac{1}{z^2}$$
$$= (z^2 + \frac{1}{z^2}) - 2$$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$$\left(z-\frac{1}{z}\right)^2 = 2\cos 2\theta - 2$$

So $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ gives

$$(2\mathrm{i}\sin\theta)^2 = 2\cos 2\theta - 2$$
$$-4\sin\theta = 2\cos 2\theta - 2$$
$$\therefore \sin^2\theta = \frac{1}{2} - \frac{1}{2}\cos 2\theta \quad \text{as req.}$$

10 a)

Since $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ and $\mathrm{e}^{-\mathrm{i}\theta} = \cos\theta - \mathrm{i}\sin\theta$

$$\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta} = 2\cos\theta$$

So $\cos\theta = \frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})$

$$\therefore \cos^3\theta = \left[\frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})\right]^3$$
$$= \frac{1}{8}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})^3$$
$$= \frac{1}{8}(\mathrm{e}^{3\mathrm{i}\theta} + 3\mathrm{e}^{-\mathrm{i}\theta} + 3\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-3\mathrm{i}\theta})$$
$$= \frac{1}{8}(\mathrm{e}^{3\mathrm{i}\theta} + \mathrm{e}^{-3\mathrm{i}\theta} + 3(\mathrm{e}^{-\mathrm{i}\theta} + \mathrm{e}^{\mathrm{i}\theta}))$$
$$= \frac{1}{8}(2\cos 3\theta + 3(2\cos\theta))$$
$$= \frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta$$

10 b)

$$\int_0^{\frac{\pi}{3}} \cos^3\theta\,\mathrm{d}x = \int_0^{\frac{\pi}{3}} \left(\frac{1}{4}\cos(3\theta) + \frac{3}{4}\cos\theta\right)\,\mathrm{d}x$$
$$= \left[\frac{1}{12}\sin(3\theta) + \frac{3}{4}\sin\theta\right]_0^{\frac{\pi}{3}}$$
$$= \left(0 + \frac{3\sqrt{3}}{8}\right) - (0+0)$$
$$= \frac{3\sqrt{3}}{8}$$

So $p = 3$ and $q = 8$

11 a)

$$\left(z-\frac{1}{z}\right)^3 = z^3 - 3z + \frac{3}{z} - \frac{1}{z^3}$$
$$= (z^3 - \frac{1}{z^3}) - 3\left(z-\frac{1}{z}\right)$$

Since $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ then

$$\left(z+\frac{1}{z}\right)^3 = 2\mathrm{i}\sin 3\theta - 3\times 2\mathrm{i}\sin\theta$$

So $(2\mathrm{i}\sin\theta)^3 = 2\mathrm{i}\sin 3\theta - 6\mathrm{i}\sin\theta$

$$-8\mathrm{i}\sin^3\theta = 2\mathrm{i}\sin 3\theta - 6\mathrm{i}\sin\theta$$
$$\therefore \sin^3\theta = -\frac{1}{4}\sin 3\theta + \frac{3}{4}\sin\theta \quad \text{as req.}$$

11 b)

$$\sin^3\theta = \frac{1}{8} \Rightarrow \sin\theta = \frac{1}{2}$$

So $\theta = \frac{\pi}{6}$ or $\theta = \frac{5\pi}{6}$

12 a)

$$\left(z-\frac{1}{z}\right)^4 = z^4 - 4z^2 + 6 - \frac{4}{z^2} + \frac{1}{z^4}$$
$$= (z^4 + \frac{1}{z^4}) - 4(z^2 + \frac{1}{z^2}) + 6$$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$$\left(z-\frac{1}{z}\right)^4 = 2\cos 4\theta - 4\times 2\cos 2\theta + 6$$

Since $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ then So $(2\mathrm{i}\sin\theta)^4 = 2\cos 4\theta - 8\cos 2\theta + 6$

$$16\sin^4\theta = 2\cos 4\theta - 8\cos 2\theta + 6$$
$$\therefore \sin^4\theta = \frac{1}{8}\cos 4\theta - \frac{1}{2}\cos 2\theta + \frac{3}{8} = \frac{1}{8}(\cos 4\theta - 4\cos 2\theta + 3)$$

So $a = 1$, $b = -4$ and $c = 3$

12 b)

Let $\theta = \frac{\pi}{8}$ so

$$\sin^4\left(\frac{\pi}{8}\right) = \frac{1}{8}\left(\cos\left(4\times\frac{\pi}{8}\right) - 4\cos\left(2\times\frac{\pi}{8}\right) + 3\right)$$
$$= \frac{1}{8}\left(\cos\left(\frac{\pi}{2}\right) - 4\cos\left(\frac{\pi}{4}\right) + 3\right)$$
$$= \frac{1}{8}\left(0 - 4\times\frac{\sqrt{2}}{2} + 3\right)$$

$$= \frac{1}{8}(3 - 2\sqrt{2})$$

$$\therefore \sin\left(\frac{\pi}{8}\right) = \left(\frac{3 - 2\sqrt{2}}{8}\right)^{\frac{1}{4}} \quad \text{as req.}$$

13 a)

$$\left(z + \frac{1}{z}\right)^4 = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$$
$$= (z^4 + \frac{1}{z^4}) + 4(z^2 + \frac{1}{z^2}) + 6$$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$$\left(z + \frac{1}{z}\right)^4 = 2\cos 4\theta + 4 \times 2\cos 2\theta + 6$$

So $(2\cos\theta)^4 = 2\cos 4\theta + 8\cos 2\theta + 6$

$$\therefore \cos^4\theta = \frac{1}{8}\cos 4\theta + \frac{1}{2}\cos 2\theta + \frac{3}{8}$$

13 b)

$$\int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} 8\cos^4\theta \,\mathrm{d}\theta = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (\cos 4\theta + 4\cos 2\theta + 3)\,\mathrm{d}\theta$$
$$= \left[\frac{1}{4}\sin 4\theta + 2\sin 2\theta + 3\theta\right]_{\frac{\pi}{2}}^{\frac{3\pi}{2}}$$
$$= \left(0 + 0 + \frac{9\pi}{2}\right) - \left(0 + 0 + \frac{3\pi}{2}\right)$$
$$= 3\pi$$

14 a)

He incorrectly defines $2\sin\theta$ as $\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}$ but $\sin\theta = \frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})$

14 b)

$$16\sin^4\theta \equiv (2\sin\theta)^4$$
$$\equiv \left(2 \times \frac{1}{2\mathrm{i}}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})\right)^4$$
$$\equiv (-\mathrm{i}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta}))^4$$
$$\equiv (-\mathrm{i})^4(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})^4$$
$$\equiv \mathrm{e}^{4\mathrm{i}\theta} - 4\mathrm{e}^{2\mathrm{i}\theta} + 6 - 4\mathrm{e}^{-2\mathrm{i}\theta} + \mathrm{e}^{-4\mathrm{i}\theta}$$
$$\equiv \mathrm{e}^{4\mathrm{i}\theta} + \mathrm{e}^{-4\mathrm{i}\theta} - 4(\mathrm{e}^{2\mathrm{i}\theta} + \mathrm{e}^{-2\mathrm{i}\theta}) + 6$$
$$\equiv 2\cos 4\theta - 8\cos 2\theta + 6$$

So that $32\sin^4\theta \equiv 4\cos 4\theta - 16\cos 2\theta + 12$ as req.

14 c)

$$\begin{aligned}16\cos^4\theta &\equiv (2\cos\theta)^4\\ &\equiv (\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})^4\\ &\equiv \mathrm{e}^{4\mathrm{i}\theta}+4\mathrm{e}^{2\mathrm{i}\theta}+6+4\mathrm{e}^{-2\mathrm{i}\theta}+\mathrm{e}^{-4\mathrm{i}\theta}\\ &\equiv \mathrm{e}^{4\mathrm{i}\theta}+\mathrm{e}^{-4\mathrm{i}\theta}+4(\mathrm{e}^{2\mathrm{i}\theta}+\mathrm{e}^{-2\mathrm{i}\theta})+6\\ &\equiv 2\cos 4\theta+8\cos 2\theta+6\end{aligned}$$

So that $32\cos^4\theta \equiv 4\cos 4\theta + 16\cos 2\theta + 12$ as req.

15 a)

Since $\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta$ and $\mathrm{e}^{-\mathrm{i}\theta} = \cos\theta - \mathrm{i}\sin\theta$

$$\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta} = 2\cos\theta$$

So $\cos\theta = \frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})$

$$\begin{aligned}\therefore \cos^5\theta &= \left[\frac{1}{2}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})\right]^5\\ &= \frac{1}{32}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})^5\\ &= \frac{1}{32}(\mathrm{e}^{5\mathrm{i}\theta}+5\mathrm{e}^{4\mathrm{i}\theta}\mathrm{e}^{-\mathrm{i}\theta}+10\mathrm{e}^{3\mathrm{i}\theta}\mathrm{e}^{-2\mathrm{i}\theta}+10\mathrm{e}^{2\mathrm{i}\theta}\mathrm{e}^{-3\mathrm{i}\theta}+5\mathrm{e}^{\mathrm{i}\theta}\mathrm{e}^{-4\mathrm{i}\theta}+\mathrm{e}^{-5\mathrm{i}\theta})\\ &= \frac{1}{32}(\mathrm{e}^{5\mathrm{i}\theta}+\mathrm{e}^{-5\mathrm{i}\theta}+5(\mathrm{e}^{3\mathrm{i}\theta}+\mathrm{e}^{-3\mathrm{i}\theta})+10(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta}))\\ &= \frac{1}{32}(2\cos 5\theta+5(2\cos 3\theta)+10(2\cos\theta))\\ &= \frac{1}{16}\cos 5\theta+\frac{5}{16}\cos 3\theta+\frac{5}{8}\cos\theta\end{aligned}$$

15 b)

$$\begin{aligned}2\cos 5\theta + 10\cos 3\theta + 20\cos\theta &= 1\\ \Rightarrow \frac{1}{16}\cos 5\theta + \frac{5}{16}\cos 3\theta + \frac{5}{8}\cos\theta &= \frac{1}{32}\\ \Rightarrow \cos^5\theta &= \frac{1}{32}\\ \Rightarrow \cos\theta &= \left(\frac{1}{32}\right)^{\frac{1}{5}}\\ \therefore \theta = \frac{\pi}{3} \text{ or } \theta &= \frac{5\pi}{3}\end{aligned}$$

16 a)

$$\begin{aligned}\left(z-\frac{1}{z}\right)^5 &= z^5 - 5z^3 + 10z - \frac{10}{z} + \frac{5}{z^3} - \frac{1}{z^5}\\ &= (z^5 - \frac{1}{z^5}) - 5(z^3 - \frac{1}{z^3}) + 10\left(z - \frac{1}{z}\right)\end{aligned}$$

Since $z^n + z^{-n} = 2\mathrm{i}\sin n\theta$ then

$$\left(z - \frac{1}{z}\right)^5 = 2\mathrm{i}\sin 5\theta - 5 \times 2\mathrm{i}\sin 3\theta + 10 \times 2\mathrm{i}\sin\theta$$

So $(2\mathrm{i}\sin\theta)^5 = 2\mathrm{i}\sin 5\theta - 10\mathrm{i}\sin 3\theta + 20\mathrm{i}\sin\theta$

$$32\mathrm{i}\sin^5\theta = 2\mathrm{i}\sin 5\theta - 10\mathrm{i}\sin 3\theta + 20\mathrm{i}\sin\theta$$
$$\therefore \sin^5\theta = \frac{1}{16}\sin 5\theta - \frac{5}{16}\sin 3\theta + \frac{5}{8}\sin\theta$$

16 b)

$$\begin{aligned}\int_0^{\frac{\pi}{2}} 16\sin^5\theta\,\mathrm{d}\theta &= \int_0^{\frac{\pi}{2}} (\sin 5\theta - 5\sin 3\theta + 10\sin\theta)\,\mathrm{d}\theta \\ &= \left[-\frac{1}{5}\cos 5\theta + \frac{5}{3}\cos 3\theta - 10\cos\theta\right]_0^{\frac{\pi}{2}} \\ &= (0+0+0) - \left(-\frac{1}{5} + \frac{5}{3} - 10\right) \\ &= \frac{128}{15}\end{aligned}$$

17 a)

$$\begin{aligned}\left(z + \frac{1}{z}\right)^6 &= z^6 + 6z^4 + 15z^2 + 20 + \frac{15}{z^2} + \frac{6}{z^4} + \frac{1}{z^6} \\ &= (z^6 + \frac{1}{z^6}) + 6(z^4 + \frac{1}{z^4}) + 15(z^2 + \frac{1}{z^2}) + 20\end{aligned}$$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$$\left(z + \frac{1}{z}\right)^6 = 2\cos 6\theta + 6 \times 2\cos 4\theta + 15 \times 2\cos 2\theta + 20$$

So $(2\cos\theta)^6 = 2\cos 6\theta + 12\cos 4\theta + 30\cos 2\theta + 20$

$$\therefore \cos^6\theta = \frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta + \frac{15}{32}\cos 2\theta + \frac{5}{16}$$

17 b)

Let $\theta = \frac{\pi}{12}$ so

$$\begin{aligned}\cos^6\left(\frac{\pi}{12}\right) &= \frac{1}{32}\cos\left(6 \times \frac{\pi}{12}\right) + \frac{3}{16}\cos\left(4 \times \frac{\pi}{12}\right) + \frac{15}{32}\cos\left(2 \times \frac{\pi}{12}\right) + \frac{5}{16} \\ &= \frac{1}{32}\cos\frac{\pi}{2} + \frac{3}{16}\cos\frac{\pi}{3} + \frac{15}{32}\cos\frac{\pi}{6} + \frac{5}{16} \\ &= 0 + \frac{3}{32} + \frac{15\sqrt{3}}{64} + \frac{5}{16} \\ &= \frac{13}{32} + \frac{15\sqrt{3}}{64} \\ &= \frac{26 + 15\sqrt{3}}{64}\end{aligned}$$

$$\therefore \cos\left(\frac{\pi}{12}\right) = \left(\frac{26 + 15\sqrt{3}}{64}\right)^{\frac{1}{6}} = \frac{1}{2}(26 + 15\sqrt{3})^{\frac{1}{6}} \quad \text{as req.}$$

18 a)

Since $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$

$$e^{i\theta} - e^{-i\theta} = 2i\sin\theta$$

So $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$

$$\begin{aligned}
\therefore \sin^6\theta &= \left[\frac{1}{2i}(e^{i\theta} - e^{-i\theta})\right]^6 \\
&= -\frac{1}{64}(e^{i\theta} - e^{-i\theta})^6 \\
&= -\frac{1}{64}(e^{6i\theta} - 6e^{5i\theta}e^{-i\theta} + 15e^{4i\theta}e^{-2i\theta} - 20e^{3i\theta}e^{-3i\theta} + 15e^{2i\theta}e^{-4i\theta} - 6e^{i\theta}e^{-5i\theta} + e^{-6i\theta}) \\
&= -\frac{1}{64}(e^{6i\theta} + e^{-6i\theta} - 6(e^{4i\theta} + e^{-4i\theta}) + 15(e^{2i\theta} + e^{-2i\theta}) - 20) \\
&= -\frac{1}{64}(2\cos 6\theta - 6(2\cos 4\theta) + 15(2\cos 2\theta) - 20) \\
&= -\frac{1}{32}\cos 6\theta + \frac{3}{16}\cos 4\theta - \frac{15}{32}\cos 2\theta + \frac{5}{16}
\end{aligned}$$

$$\therefore 32\sin^6\theta = -\cos 6\theta + 6\cos 4\theta - 15\cos 2\theta + 10$$

18 b)

Let $\theta = \frac{\pi}{12}$ so

$$\begin{aligned}
\sin^6\left(\frac{\pi}{12}\right) &= -\frac{1}{32}\cos\left(6\times\frac{\pi}{12}\right) + \frac{3}{16}\cos\left(4\times\frac{\pi}{12}\right) - \frac{15}{32}\cos\left(2\times\frac{\pi}{12}\right) + \frac{5}{16} \\
&= -\frac{1}{32}\cos\left(\frac{\pi}{2}\right) + \frac{3}{16}\cos\left(\frac{\pi}{3}\right) - \frac{15}{32}\cos\left(\frac{\pi}{6}\right) + \frac{5}{16} \\
&= -\frac{1}{32}\cos\frac{\pi}{2} + \frac{3}{16}\cos\frac{\pi}{3} - \frac{15}{32}\cos\frac{\pi}{6} + \frac{5}{16} \\
&= 0 + \frac{3}{32} - \frac{15\sqrt{3}}{64} + \frac{5}{16} \\
&= \frac{13}{32} - \frac{15\sqrt{3}}{64} \\
&= \frac{26 - 15\sqrt{3}}{64}
\end{aligned}$$

$$\therefore \sin\left(\frac{\pi}{12}\right) = \left(\frac{26 - 15\sqrt{3}}{64}\right)^{\frac{1}{6}} = \frac{1}{2}(26 - 15\sqrt{3})^{\frac{1}{6}} \quad \text{as req.}$$

19)

$$\begin{aligned}
\left(z^3 + \frac{1}{z^3}\right)^2 &= z^6 + 2 + \frac{1}{z^6} \\
&= (z^6 + \frac{1}{z^6}) + 2
\end{aligned}$$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$$\left(z^3 + \frac{1}{z^3}\right)^2 = 2\cos 6\theta + 2$$

But also

$$\left(z^3+\frac{1}{z^3}\right)^2=(2\cos 3\theta)^2=4\cos^2 3\theta$$

So $4\cos^2 3\theta=2\cos 6\theta+2$

$$\therefore 2\cos^2 3\theta=\cos 6\theta+1$$

20)

$$\begin{aligned}\left(z^4-\frac{1}{z^4}\right)^3&=z^{12}-3z^4+\frac{3}{z^4}-\frac{1}{z^{12}}\\&=(z^{12}-\frac{1}{z^{12}})-3\left(z^4-\frac{1}{z^4}\right)\end{aligned}$$

Since $z^n-z^{-n}=2\mathrm{i}\sin n\theta$ then

$$\left(z^4-\frac{1}{z^4}\right)^3=2\mathrm{i}\sin 12\theta-6\mathrm{i}\sin 4\theta$$

But also

$$\left(z^4-\frac{1}{z^4}\right)^3=(2\mathrm{i}\sin 4\theta)^3=-8\mathrm{i}\sin^3 4\theta$$

So $-8\mathrm{i}\sin^3 4\theta=2\mathrm{i}\sin 12\theta-6\mathrm{i}\sin 4\theta$

$$\therefore 4\sin^3 4\theta=-\sin 12\theta+3\sin 4\theta\quad\text{as req.}$$

21)

$$\begin{aligned}\left(z^3-\frac{1}{z^3}\right)^5&=z^{15}-5z^9+10z^3-\frac{10}{z^3}+\frac{5}{z^9}-\frac{1}{z^{15}}\\&=(z^{15}-\frac{1}{z^{15}})-5(z^9-\frac{1}{z^9})+10\left(z^3-\frac{1}{z^3}\right)\end{aligned}$$

Since $z^n-z^{-n}=2\mathrm{i}\sin n\theta$ then

$$\left(z^3-\frac{1}{z^3}\right)^5=2\mathrm{i}\sin 15\theta-10\mathrm{i}\sin 9\theta+20\mathrm{i}\sin 3\theta$$

But also

$$\left(z^3-\frac{1}{z^3}\right)^5=(2\mathrm{i}\sin 3\theta)^5=32\mathrm{i}\sin^5 3\theta$$

So $32\mathrm{i}\sin^5 3\theta=2\mathrm{i}\sin 15\theta-10\mathrm{i}\sin 9\theta+20\mathrm{i}\sin 3\theta$

$$\therefore \sin^5 3\theta=\frac{1}{16}\sin 15\theta-\frac{5}{16}\sin 9\theta+\frac{5}{8}\sin 3\theta$$

22 a)

$$z^n=\cos n\theta+\mathrm{i}\sin n\theta$$

$$z^{-n}=\cos(-n\theta)+\mathrm{i}\sin(-n\theta)=\cos n\theta-\mathrm{i}\sin n\theta$$

$$z^n+z^{-n}=\cos n\theta+\mathrm{i}\sin n\theta+\cos n\theta-\mathrm{i}\sin n\theta=2\cos n\theta$$

22 b)

$$z^n - z^{-n} = \cos n\theta + \mathrm{i}\sin n\theta - \cos n\theta + \mathrm{i}\sin n\theta = 2\mathrm{i}\sin n\theta$$

22 c)

$$\begin{aligned}\left(z+\frac{1}{z}\right)^2\left(1-\frac{1}{z}\right)^2 &= \left(z^2+1+1+\frac{1}{z^2}\right)\left(z^2-1-1+\frac{1}{z^2}\right)\\ &= \left(z^2+2+\frac{1}{z^2}\right)\left(z^2-2+\frac{1}{z^2}\right)\\ &= z^4-2z^2+1+2z^2-4+\frac{2}{z^2}+1-\frac{2}{z^2}+\frac{1}{z^4}\\ &= \left(z^4+\frac{1}{z^4}\right)-2\end{aligned}$$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$$\left(z+\frac{1}{z}\right)^2\left(1-\frac{1}{z}\right)^2 = 2\cos 4\theta - 2$$

But also

$$\left(z+\frac{1}{z}\right)^2\left(1-\frac{1}{z}\right)^2 = (2\cos\theta)^2(2\mathrm{i}\sin\theta)^2 = -16\cos^2\theta\sin^2\theta$$

So $-16\cos^2\theta\sin^2\theta = 2\cos 4\theta - 2$

$$\therefore \cos^2\theta\sin^2\theta = -\frac{2}{16}\cos 4\theta + \frac{2}{16} = \frac{1}{8}(1-\cos 4\theta) \quad \text{as req.}$$

23 a)

$$\begin{aligned}z^n &= \cos n\theta + \mathrm{i}\sin n\theta\\ z^{-n} &= \cos(-n\theta) + \mathrm{i}\sin(-n\theta) = \cos n\theta - \mathrm{i}\sin n\theta\\ z^n + z^{-n} &= \cos n\theta + \mathrm{i}\sin n\theta + \cos n\theta - \mathrm{i}\sin n\theta = 2\cos n\theta\end{aligned}$$

23 b)

$$z^n - z^{-n} = \cos n\theta + \mathrm{i}\sin n\theta - \cos n\theta + \mathrm{i}\sin n\theta = 2\mathrm{i}\sin n\theta$$

23 c)

$$\begin{aligned}&\left(z+\frac{1}{z}\right)^3\left(z-\frac{1}{z}\right)^3\\ &= \left(z^3+3z+\frac{3}{z}+\frac{1}{z^3}\right)\left(z^3-3z+\frac{3}{z}-\frac{1}{z^3}\right)\\ &= z^6-3z^4+3z^2-1+3z^4-9z^2+9-\frac{3}{z^2}+3z^2-9+\frac{9}{z^2}-\frac{3}{z^4}+1-\frac{3}{z^2}+\frac{3}{z^4}-\frac{1}{z^6}\\ &= \left(z^6-\frac{1}{z^6}\right)-3\left(z^2-\frac{3}{z^2}\right)\end{aligned}$$

Since $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ then

$$\left(z+\frac{1}{z}\right)^3\left(1-\frac{1}{z}\right)^3 = 2\mathrm{i}\sin 6\theta - 6\mathrm{i}\sin 2\theta$$

But also

$$\left(z+\frac{1}{z}\right)^3\left(1-\frac{1}{z}\right)^3 = (2\cos\theta)^3(2\mathrm{i}\sin\theta)^3 = -64\mathrm{i}\cos^3\theta\sin^3\theta$$

So $-64\mathrm{i}\cos^3\theta\sin^3\theta = 2\mathrm{i}\sin 6\theta - 6\mathrm{i}\sin 2\theta$

$$\therefore \cos^3\theta\sin^3\theta = -\frac{1}{32}\sin 6\theta + \frac{3}{32}\sin 2\theta = \frac{1}{32}(3\sin 2\theta - \sin 6\theta) \quad \text{as req.}$$

24)

Let $z = \cos\theta + \mathrm{i}\sin\theta$

$$\begin{aligned}\left(z+\frac{1}{z}\right)^3\left(z-\frac{1}{z}\right)^4 &= \left(z^3+3z+\frac{3}{z}+\frac{1}{z^3}\right)\left(z^4-4z^2+6-\frac{4}{z^2}+\frac{1}{z^4}\right)\\ &= z^7+3z^5+3z^3+z+-4z^5-12z^3-12z-\frac{4}{z}+6z^3+18z\\ &\quad+\frac{18}{z}+\frac{6}{z^3}-\frac{4}{z^5}-\frac{12}{z^3}-4z-\frac{12}{z}+\frac{1}{z}+\frac{3}{z^3}+\frac{3}{z^5}+\frac{1}{z^7}\\ &= \left(z^7+\frac{1}{z^7}\right)+\left(-z^5-\frac{1}{z^5}\right)+\left(-3z^3-\frac{3}{z^3}\right)+\left(3z+\frac{3}{z}\right)\\ &= \left(z^7+\frac{1}{z^7}\right)-\left(z^5+\frac{1}{z^5}\right)-\left(3z^3+\frac{3}{z^3}\right)+\left(3z+\frac{3}{z}\right)\end{aligned}$$

So that since $z^n + z^{-n} = 2\cos n\theta$ and $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ then

$$\begin{aligned}(2\cos\theta)^3(2\mathrm{i}\sin\theta)^4 &= 2\cos 7\theta - 2\cos 5\theta - 3\times 2\cos 3\theta + 3\times 2\cos\theta\\ 8\cos^3\theta\times 16\sin^4\theta &= 2\cos 7\theta - 2\cos 5\theta - 6\cos 3\theta + 6\cos\theta\\ \cos^3\theta\sin^4\theta &= \frac{1}{64}\cos 7\theta - \frac{1}{64}\cos 5\theta - \frac{3}{64}\cos 3\theta + \frac{3}{64}\cos\theta\\ &= \frac{1}{64}(\cos 7\theta - \cos 5\theta - 3\cos 3\theta + 3\cos\theta)\end{aligned}$$

25 a)

$$\begin{aligned}\mathrm{Re}\left(\mathrm{e}^{3\mathrm{i}\theta}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})^3\right) &= \mathrm{Re}((\cos 3\theta + \mathrm{i}\sin 3\theta)(2\cos\theta)^3)\\ &= 8\cos 3\theta\cos^3\theta \quad \text{so } k = 8\end{aligned}$$

25 b)

$$\begin{aligned}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})^3 &= \mathrm{e}^{3\mathrm{i}\theta}+3\mathrm{e}^{\mathrm{i}\theta}+3\mathrm{e}^{-\mathrm{i}\theta}+\mathrm{e}^{-3\mathrm{i}\theta}\\ \therefore \mathrm{e}^{3\mathrm{i}\theta}(\mathrm{e}^{\mathrm{i}\theta}+\mathrm{e}^{-\mathrm{i}\theta})^3 &= \mathrm{e}^{3\mathrm{i}\theta}(\mathrm{e}^{3\mathrm{i}\theta}+3\mathrm{e}^{\mathrm{i}\theta}+3\mathrm{e}^{-\mathrm{i}\theta}+\mathrm{e}^{-3\mathrm{i}\theta})\\ &= \mathrm{e}^{6\mathrm{i}\theta}+3\mathrm{e}^{4\mathrm{i}\theta}+3\mathrm{e}^{2\mathrm{i}\theta}+1\end{aligned}$$

So $\mathrm{Re}(\mathrm{e}^{6\mathrm{i}\theta}+3\mathrm{e}^{4\mathrm{i}\theta}+3\mathrm{e}^{2\mathrm{i}\theta}+1) = \cos 6\theta + 3\cos 4\theta + 3\cos 2\theta + 1$

$$\therefore 8\cos 3\theta\cos^3\theta = \cos 6\theta + 3\cos 4\theta + 3\cos 2\theta + 1$$

Let $\theta = \frac{\pi}{12}$

$$8\cos\frac{\pi}{4}\cos^3\frac{\pi}{12} = \cos\frac{\pi}{2} + 3\cos\frac{\pi}{3} + 3\cos\frac{\pi}{6} + 1$$
$$\Rightarrow 4\sqrt{2}\cos^3\frac{\pi}{12} = 0 + \frac{3}{2} + \frac{3\sqrt{3}}{2} + 1$$
$$\Rightarrow \cos^3\frac{\pi}{12} = \frac{5}{8\sqrt{2}} + \frac{3\sqrt{3}}{8\sqrt{2}} = \frac{5\sqrt{2}+3\sqrt{6}}{16} = \frac{10\sqrt{2}}{32} + \frac{3\sqrt{6}}{32}$$
$$\therefore \cos\frac{\pi}{12} = \sqrt[3]{\frac{5\sqrt{2}+3\sqrt{6}}{16}}$$

So $p = 5$ and $q = 3$

26 a)

$$\begin{aligned}\operatorname{Im}\left(\mathrm{e}^{4\mathrm{i}\theta}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})^4\right) &= \operatorname{Im}((\cos 4\theta + \mathrm{i}\sin 4\theta)(2\mathrm{i}\sin\theta)^4) \\ &= 16\sin 4\theta \sin^4\theta \quad \text{so } k = 16\end{aligned}$$

26 b)

$$\begin{aligned}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})^4 &= \mathrm{e}^{4\mathrm{i}\theta} - 4\mathrm{e}^{2\mathrm{i}\theta} + 6 - 4\mathrm{e}^{-2\mathrm{i}\theta} + \mathrm{e}^{-4\mathrm{i}\theta} \\ \therefore \mathrm{e}^{4\mathrm{i}\theta}(\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta})^4 &= \mathrm{e}^{4\mathrm{i}\theta}(\mathrm{e}^{4\mathrm{i}\theta} - 4\mathrm{e}^{2\mathrm{i}\theta} + 6 - 4\mathrm{e}^{-2\mathrm{i}\theta} + \mathrm{e}^{-4\mathrm{i}\theta}) \\ &= \mathrm{e}^{8\mathrm{i}\theta} - 4\mathrm{e}^{6\mathrm{i}\theta} + 6\mathrm{e}^{4\mathrm{i}\theta} - 4\mathrm{e}^{2\mathrm{i}\theta} + 1\end{aligned}$$

So

$$\operatorname{Im}(\mathrm{e}^{8\mathrm{i}\theta} - 4\mathrm{e}^{6\mathrm{i}\theta} + 6\mathrm{e}^{4\mathrm{i}\theta} - 4\mathrm{e}^{2\mathrm{i}\theta}) = \sin 8\theta - 4\sin 6\theta + 6\sin 4\theta - 4\sin 2\theta$$
$$\therefore 16\sin 4\theta\sin^4\theta = \sin 8\theta - 4\sin 6\theta + 6\sin 4\theta - 4\sin 2\theta$$

Let $\theta = \frac{\pi}{8}$

$$16\sin\frac{\pi}{2}\sin^4\frac{\pi}{8} = \sin\pi - 4\sin\frac{3\pi}{4} + 6\sin\frac{\pi}{2} - 4\sin\frac{\pi}{4}$$
$$\Rightarrow 16\sin^4\frac{\pi}{8} = 0 - 2\sqrt{2} + 6 - 2\sqrt{2}$$
$$\Rightarrow \sin^4\frac{\pi}{8} = \frac{6-4\sqrt{2}}{16}$$
$$\therefore \sin\frac{\pi}{8} = \sqrt[4]{\frac{6-4\sqrt{2}}{16}} = \frac{1}{2}(6-4\sqrt{2})^{\frac{1}{4}}$$

So $p = 6$ and $q = 4$

27 a)

$$\int \sin\theta(\cos\theta)^4\,\mathrm{d}\theta = -\frac{1}{5}\cos^5\theta + C$$

27 b)

$$\begin{aligned}\left(z+\frac{1}{z}\right)^4\left(z-\frac{1}{z}\right) &= \left(z^4+4z^2+6+\frac{4}{z^2}+\frac{1}{z^4}\right)\left(z-\frac{1}{z}\right)\\ &= z^5+4z^3+6z+\frac{4}{z}+\frac{1}{z^3}-z^3-4z-\frac{6}{z}-\frac{4}{z^3}-\frac{1}{z^5}\\ &= \left(z^5-\frac{1}{z^5}\right)+3\left(z^3-\frac{1}{z^3}\right)+2\left(z-\frac{1}{z}\right)\end{aligned}$$

Since $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ then

$$\left(z+\frac{1}{z}\right)^4\left(z-\frac{1}{z}\right) = 2\mathrm{i}\sin 5\theta + 6\mathrm{i}\sin 3\theta + 4\mathrm{i}\sin\theta$$

But also

$$\left(z+\frac{1}{z}\right)^4\left(z-\frac{1}{z}\right) = (2\cos\theta)^4 2\mathrm{i}\sin\theta = 32\mathrm{i}\cos^4\theta\sin\theta$$

So $32\mathrm{i}\cos^4\theta\sin\theta = 2\mathrm{i}\sin 5\theta + 6\mathrm{i}\sin 3\theta + 4\mathrm{i}\sin\theta$

$$\therefore \cos^4\theta\sin\theta = \frac{1}{16}\sin 5\theta + \frac{3}{16}\sin 3\theta + \frac{1}{8}\sin\theta$$

$$\begin{aligned}\int \sin\theta(\cos\theta)^4\,\mathrm{d}\theta &= \int\left(\frac{1}{16}\sin 5\theta + \frac{3}{16}\sin 3\theta + \frac{1}{8}\sin\theta\right)\mathrm{d}\theta\\ &= -\frac{1}{80}\cos 5\theta - \frac{1}{16}\cos 3\theta - \frac{1}{8}\cos\theta + C\end{aligned}$$

28 a)

$$\begin{aligned}\left(z-\frac{1}{z}\right)^3\left(z+\frac{1}{z}\right)^2 &= \left(z^3-3z+\frac{3}{z}-\frac{1}{z^3}\right)\left(z^2+2+\frac{1}{z^2}\right)\\ &= z^5+2z^3+z-3z^3-6z-\frac{3}{z}+3z+\frac{6}{z}+\frac{3}{z^3}-\frac{1}{z}-\frac{2}{z^3}-\frac{1}{z^5}\\ &= \left(z^5-\frac{1}{z^5}\right)-\left(z^3-\frac{1}{z^3}\right)-2\left(z-\frac{1}{z}\right)\end{aligned}$$

Since $z^n - z^{-n} = 2\mathrm{i}\sin n\theta$ then

$$\left(z-\frac{1}{z}\right)^3\left(z+\frac{1}{z}\right)^2 = 2\mathrm{i}\sin 5\theta - 2\mathrm{i}\sin 3\theta - 4\mathrm{i}\sin\theta$$

But also

$$\left(z-\frac{1}{z}\right)^3\left(z+\frac{1}{z}\right)^2 = (2\mathrm{i}\sin\theta)^3(2\cos\theta)^2 = -32\mathrm{i}\sin^3\theta\cos^2\theta$$

So $-32\mathrm{i}\sin^3\theta\cos^2\theta = 2\mathrm{i}\sin 5\theta - 2\mathrm{i}\sin 3\theta - 4\mathrm{i}\sin\theta$

$$\therefore 16\sin^3\theta\cos^2\theta = 2\sin\theta + \sin 3\theta - \sin 5\theta$$

So $a = 2$, $b = 1$ and $c = -1$

28 b)

Choose $x = \sin\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \cos\theta$ and when $x = 0$, $\theta = 0$, when $x = 1$, $\theta = \frac{\pi}{2}$

$$\begin{aligned}\int_0^{\frac{\pi}{2}} 16\sin^3\theta\sqrt{1-\sin^2\theta}\times\cos\theta\,\mathrm{d}\theta &= \int_0^{\frac{\pi}{2}} 16\sin^3\theta\cos^2\theta\,\mathrm{d}\theta \\ &= \int_0^{\frac{\pi}{2}} (2\sin\theta + \sin 3\theta - \sin 5\theta)\,\mathrm{d}\theta \\ &= \left[-2\cos\theta - \frac{1}{3}\cos 3\theta + \frac{1}{5}\cos 5\theta\right]_0^{\frac{\pi}{2}} \\ &= (0+0+0) - \left(-2 - \frac{1}{3} + \frac{1}{5}\right) = \frac{32}{15} \quad \text{as req.}\end{aligned}$$

29 a)

$$\begin{aligned}\left(z-\frac{1}{z}\right)^2\left(z+\frac{1}{z}\right)^2 &= \left(z^2 - 2 + \frac{1}{z^2}\right)\left(z^2 + 2 + \frac{1}{z^2}\right) \\ &= z^4 - 2z^2 + 1 + 2z^2 - 4 + \frac{2}{z^2} + 1 - \frac{2}{z^2} + \frac{1}{z^4} \\ &= \left(z^4 + \frac{1}{z^4}\right) - 2\end{aligned}$$

Since $z^n - z^{-n} = 2\cos n\theta$

$$\left(z-\frac{1}{z}\right)^2\left(z+\frac{1}{z}\right)^2 = 2\cos 4\theta - 2$$

But also

$$\left(z-\frac{1}{z}\right)^2\left(z+\frac{1}{z}\right)^2 = (2\mathrm{i}\sin\theta)^2(2\cos\theta)^2 = -16\sin^2\theta\cos^2\theta$$

So $-16\sin^2\theta\cos^2\theta = 2\cos 4\theta - 2$

$$\therefore 8\sin^2\theta\cos^2\theta = 1 - \cos 4\theta$$

So $a = 1$ and $b = -1$

29 b)

Choose $x = 2\sin\theta$

$\frac{\mathrm{d}x}{\mathrm{d}\theta} = 2\cos\theta$ and when $x = 0$, $\theta = 0$, when $x = 2$, $\theta = \frac{\pi}{2}$

$$\begin{aligned}\int_0^{\frac{\pi}{2}} 4\times 4\sin^2\theta\sqrt{4-4\sin^2\theta}\times 2\cos\theta\,\mathrm{d}\theta &= \int_0^{\frac{\pi}{2}} 32\sin^2\theta\times 2\sqrt{1-1\sin^2\theta}\times\cos\theta\,\mathrm{d}\theta \\ &= \int_0^{\frac{\pi}{2}} 64\sin^2\theta\cos^2\theta\,\mathrm{d}\theta \\ &= \int_0^{\frac{\pi}{2}} (8 - 8\cos 4\theta)\,\mathrm{d}\theta \\ &= [8\theta - 2\sin 4\theta]_0^{\frac{\pi}{2}} \\ &= (4\pi - 0) - (0 - 0) = 4\pi \quad \text{as req.}\end{aligned}$$

30 a)

$$\begin{aligned}\left(z-\frac{1}{z}\right)^6 &= z^6-6z^4+15z^2-20+\frac{15}{z^2}-\frac{6}{z^4}+\frac{1}{z^6}\\ &= \left(z^6+\frac{1}{z^6}\right)-6\left(z^4+\frac{1}{z^4}\right)+15\left(z^2+\frac{1}{z^2}\right)-20\end{aligned}$$

Since $z^n+z^{-n}=2\cos n\theta$ then

$$\left(z-\frac{1}{z}\right)^6 = 2\cos 6\theta-6\times 2\cos 4\theta+15\times 2\cos 2\theta-20$$

So $(2\mathrm{i}\sin\theta)^6 = 2\cos 6\theta-12\cos 4\theta+30\cos 2\theta-20$

$$\begin{aligned}-64\sin^6\theta &= 2\cos 6\theta-12\cos 4\theta+30\cos 2\theta-20\\ \therefore \sin^6\theta &= -\frac{1}{32}\cos 6\theta+\frac{3}{16}\cos 4\theta-\frac{15}{32}\cos 2\theta+\frac{5}{16}\end{aligned}$$

30 b)

$$\begin{aligned}\int_0^{\frac{\pi}{4}}\sin^6\theta\,\mathrm{d}\theta &= \int_0^{\frac{\pi}{4}}\left(-\frac{1}{32}\cos 6\theta+\frac{3}{16}\cos 4\theta-\frac{15}{32}\cos 2\theta+\frac{5}{16}\right)\mathrm{d}\theta\\ &= \left[-\frac{1}{192}\sin 6\theta+\frac{3}{64}\sin 4\theta-\frac{15}{64}\sin 2\theta+\frac{5}{16}\theta\right]_0^{\frac{\pi}{4}}\\ &= \left(\frac{1}{192}+0-\frac{15}{64}+\frac{5}{16}\times\frac{\pi}{4}\right)-(0)\\ &= -\frac{11}{48}+\frac{5\pi}{64}\\ &= \frac{-44+15\pi}{192}\end{aligned}$$

$$\therefore p=15,\quad q=44 \text{ and } r=192$$

30 c)

$$\begin{aligned}I_6 &= \frac{6-1}{6}I_4-\frac{1}{6\times 2^3} = \frac{5}{6}I_4-\frac{1}{48}\\ I_4 &= \frac{4-1}{4}I_2-\frac{1}{4\times 2^2} = \frac{3}{4}I_2-\frac{1}{16}\\ I_2 &= \frac{2-1}{2}I_0-\frac{1}{2\times 2^1} = \frac{1}{2}I_0-\frac{1}{4}\end{aligned}$$

Since $I_0=\int_0^{\frac{\pi}{4}}\sin^0\theta\,\mathrm{d}\theta=\int_0^{\frac{\pi}{4}}1\,\mathrm{d}\theta=[\theta]_0^{\frac{\pi}{4}}=\frac{\pi}{4}$ then

$$\begin{aligned}I_2 &= \frac{\pi}{8}-\frac{1}{4}\\ \Rightarrow I_4 &= \frac{3}{4}\left(\frac{\pi}{8}-\frac{1}{4}\right)-\frac{1}{16} = \frac{3\pi}{24}-\frac{1}{4}\\ \Rightarrow I_6 &= \frac{5}{6}\left(\frac{3\pi}{24}-\frac{1}{4}\right)-\frac{1}{48} = \frac{15\pi}{192}-\frac{11}{48} = \frac{15\pi-44}{192}\end{aligned}$$

which is the same result as before.

31 a)

$$z^n = \cos n\theta + \mathrm{i}\sin n\theta$$
$$z^{-n} = \cos(-n\theta) + \mathrm{i}\sin(-n\theta) = \cos n\theta - \mathrm{i}\sin n\theta$$
$$z^n + z^{-n} = \cos n\theta + \mathrm{i}\sin n\theta + \cos n\theta - \mathrm{i}\sin n\theta = 2\cos n\theta$$
$$\therefore z^n + \frac{1}{z^n} = 2\cos n\theta \quad \text{as req.}$$

31 b)

Let $f(x) = 8x^3 - 12x^2 + 6x - 1$

$$f\left(\frac{1}{2}\right) = 1 - 3 + 3 - 1 = 0 \quad \therefore (2x-1) \text{ is a factor of } 8x^3 - 12x^2 + 6x - 1$$

31 c)

$$\left(z + \frac{1}{z}\right)^3 = z^3 + 3z + \frac{3}{z} + \frac{1}{z^3} = \left(z^3 + \frac{1}{z^3}\right) + 3\left(z + \frac{1}{z}\right)$$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$$\left(z + \frac{1}{z}\right)^3 = 2\cos 3\theta + 3 \times 2\cos\theta$$

So $(2\cos\theta)^3 = 2\cos 3\theta + 6\cos\theta$

$$\therefore 8\cos^3\theta = 2\cos 3\theta + 6\cos\theta \quad \text{as req.}$$

31 d)

$$z^6 - 3z^5 + 6z^4 - 7z^3 + 6z^2 - 3z + 1 = 0$$
$$(z^6 + 1) - 3(z^5 + z) + 6(z^4 + z^2) - 7z^3 = 0$$
$$\left(z^3 + \frac{1}{z^3}\right) - 3\left(z^2 + \frac{1}{z^2}\right) + 6\left(z + \frac{1}{z}\right) - 7 = 0$$
$$(2\cos 3\theta) - 3(2\cos 2\theta) + 6(2\cos\theta) - 7 = 0$$
$$2\cos 3\theta + 6\cos\theta - 6\cos 2\theta + 6\cos\theta - 7 = 0$$
$$8\cos^3\theta - 6(2\cos^2\theta - 1) + 6\cos\theta - 7 = 0$$
$$8\cos^3\theta - 12\cos^2\theta + 6 + 6\cos\theta - 7 = 0$$
$$8\cos^3\theta - 12\cos^2\theta + 6\cos\theta - 1 = 0$$
$$(2\cos\theta - 1)^3 = 0$$
$$\Rightarrow \cos\theta = \frac{1}{2}$$
$$\theta = \frac{\pi}{3} \text{ or } \theta = \frac{5\pi}{3}$$

gives $z = \cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3} = \frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$ and $z = \cos\frac{5\pi}{3} + \mathrm{i}\sin\frac{5\pi}{3} = \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

so the solutions are $\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$ and $\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

32 a)

$$\begin{aligned} z^n &= \cos n\theta + \mathrm{i}\sin n\theta \\ z^{-n} &= \cos(-n\theta) + \mathrm{i}\sin(-n\theta) = \cos n\theta - \mathrm{i}\sin n\theta \\ z^n + z^{-n} &= \cos n\theta + \mathrm{i}\sin n\theta + \cos n\theta - \mathrm{i}\sin n\theta = 2\cos n\theta \end{aligned}$$

$$\therefore z^n + \frac{1}{z^n} = 2\cos n\theta \quad \text{as req.}$$

32 b)

$$\left(z + \frac{1}{z}\right)^3 = z^3 + 3z + \frac{3}{z} + \frac{1}{z^3} = \left(z^3 + \frac{1}{z^3}\right) + 3\left(z + \frac{1}{z}\right)$$

Since $z^n + z^{-n} = 2\cos n\theta$ then

$$\left(z + \frac{1}{z}\right)^3 = 2\cos 3\theta + 3 \times 2\cos\theta$$

So $(2\cos\theta)^3 = 2\cos 3\theta + 6\cos\theta$

$$\therefore \cos^3\theta = \frac{1}{4}\cos 3\theta + \frac{3}{4}\cos\theta = \frac{1}{4}(\cos 3\theta + 3\cos\theta) \quad \text{as req.}$$

32 c)

$$\begin{gathered} z^6 + 3z^4 + z^3 + 3z^2 + 1 = 0 \\ (z^6 + 1) + 3(z^4 + z^2) + z^3 = 0 \\ \left(z^3 + \frac{1}{z^3}\right) + 3\left(z + \frac{1}{z}\right) + 1 = 0 \\ 2\cos(3\theta) + 6\cos\theta + 2 = 0 \\ 8\cos^3\theta + 1 = 0 \\ \cos^3\theta = -\frac{1}{8} \\ \cos\theta = -\frac{1}{2} \\ \theta = \frac{2\pi}{3} \text{ or } \theta = \frac{4\pi}{3} \end{gathered}$$

gives $z = \cos\frac{2\pi}{3} + \mathrm{i}\sin\frac{2\pi}{3} = -\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}$ and $z = \cos\frac{4\pi}{3} + \mathrm{i}\sin\frac{4\pi}{3} = -\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$

$$\left(z - \left(-\frac{1}{2} + \frac{\sqrt{3}}{2}\mathrm{i}\right)\right)\left(z - \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right)\right) = z^2 + z + 1$$

$$z^6 + 3z^4 + z^3 + 3z^2 + 1 = (z^2 + z + 1)(z^4 - z^3 + 3z^2 - z + 1)$$

33 a)

$$\left(z+\frac{1}{z}\right)^5 = z^5+5z^3+10z+\frac{10}{z}+\frac{5}{z^3}+\frac{1}{z^5}$$
$$= \left(z^5+\frac{1}{z^5}\right)+5\left(z^3+\frac{1}{z^3}\right)+10\left(z+\frac{1}{z}\right)$$

Since $z^n+z^{-n}=2\cos n\theta$ then

$$\left(z+\frac{1}{z}\right)^5 = 2\cos 5\theta+5\times 2\cos 3\theta+10\times 2\cos\theta$$

So $(2\cos\theta)^5 = 2\cos 5\theta+10\cos 3\theta+20\cos\theta$

$$\therefore 32\cos^5\theta = 2\cos 5\theta+10\cos 3\theta+20\cos\theta$$

33 b)

$$z^{10}+5z^8+10z^6-z^5+10z^4+5z^2+1=0$$
$$(z^{10}+1)+5(z^8+z^2)+10(z^6+z^4)-z^5=0$$
$$\left(z^5+\frac{1}{z^5}\right)+5\left(z^3+\frac{1}{z^3}\right)+10\left(z+\frac{1}{z}\right)-1=0$$
$$2\cos 5\theta+5(2\cos 3\theta)+10(2\cos\theta)-1=0$$
$$2\cos 5\theta+10\cos 3\theta+20\cos\theta-1=0$$
$$32\cos^5\theta=1$$
$$\cos^5\theta=\frac{1}{32}$$
$$\cos\theta=\frac{1}{2}$$
$$\theta=\frac{\pi}{3}\text{ or }\theta=\frac{2\pi}{3}$$

gives $z=\cos\frac{\pi}{3}+\mathrm{i}\sin\frac{\pi}{3}=\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}$ or $z=\cos\frac{2\pi}{3}+\mathrm{i}\sin\frac{2\pi}{3}=\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}$

$$\left(z-\left(\frac{1}{2}+\frac{\sqrt{3}}{2}\mathrm{i}\right)\right)\left(z-\left(\frac{1}{2}-\frac{\sqrt{3}}{2}\mathrm{i}\right)\right)=z^2-z+1$$

$$z^{10}+5z^8+10z^6-z^5+10z^4+5z^2+1$$
$$=(z^2-z+1)(z^8+z^7+5z^6+4z^5+9z^4+4z^3+5z^2+z+1)$$

34)

$$25z^4-10z^3+2z^2-10z+25=0$$
$$25(z^4+1)-10(z^3+z)+2z^2=0$$
$$25\left(z^2+\frac{1}{z^2}\right)-10\left(z+\frac{1}{z}\right)+2=0$$
$$25(2\cos(2\theta))-10(2\cos\theta)+2=0$$

$$25\cos(2\theta) - 10\cos\theta + 1 = 0$$
$$25(2\cos^2\theta - 1) - 10\cos\theta + 1 = 0$$
$$50\cos^2\theta - 25 - 10\cos\theta + 1 = 0$$
$$25\cos^2\theta - 5\cos\theta - 12 = 0$$
$$(5\cos\theta + 3)(5\cos\theta - 4) = 0$$

Solving gives:

$$\cos\theta = -\frac{3}{5} \Rightarrow \sin\theta = \pm\frac{4}{5}$$
$$\cos\theta = \frac{4}{5} \Rightarrow \sin\theta = \pm\frac{3}{5}$$
$$\therefore z = -\frac{3}{5} + \frac{4}{5}\text{i or } z = -\frac{3}{5} - \frac{4}{5}\text{i or } z = \frac{4}{5} - \frac{3}{5}\text{i or } z = \frac{4}{5} + \frac{3}{5}\text{i}$$